

RAHUL GULIA

Ph.D. in Electrical and Computer Engineering specializing in AI-native wireless systems, physics-informed machine learning, and digital twins for next-generation communication and sensing networks.

RESEARCH INTERESTS

- AI-enabled wireless systems and RF sensing
- Joint communication and sensing (JCAS)
- Physics-informed and interpretable machine learning
- Digital twins for cyber-physical systems
- Secure and energy-efficient wireless architectures

CONTACT

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EDUCATION

- **PhD in Electrical & Computer Engineering** | May 2026
Rochester Institute of Technology,
Advisor: Dr. Amlan Ganguly
- **M.S. in Electrical Engineering** | Aug 2020,
Rochester Institute of Technology
Master's Thesis: [Path Loss Model for 2.4 GHz Indoor Wireless Networks with Application to Drones](#)
- **B.E. in Electronics & Communication Engineering** | May 2014
Visvesvaraya Technological University

SKILLS

Wireless, RF & Sensing

- Wireless PHY/MAC, OFDM, PAPR optimization, joint communication–sensing
- Backscatter and passive sensing systems
- V2X, mmWave/60 GHz, industrial wireless

AI/ML for Physical Systems

- Physics-informed ML, VAEs, Reinforcement Learning
- Temporal models (ConvLSTM, latent memory)
- Interpretable ML & Symbolic Regression.

PUBLICATIONS

- R. Gulia et al., "AI-Driven Radio Propagation Prediction in Automated Warehouses using Variational Autoencoders," *IEEE Transactions on Industrial Informatics*, Accepted, 2026
- R. Gulia et al., "Memory-Augmented Generative AI for Real-time Wireless Prediction in Dynamic Industrial Environments," *IEEE Transactions on Industrial Informatics* [In Revision].
- R. Gulia, "[4D Fresnel Space-Time Modulation for Near-Field ELAA: Kinematic Multiplexing and \$O\(N \log N\)\$ Precoding at Sub-THz Frequencies](#)," *IEEE Transactions on Vehicular Technology*, 2026 [In Revision].
- R. Gulia et al., "Hardware-Aware Backscatter-Compatible Antenna Selection for Ultra-Reliable Industrial Wireless Sensor Networks," *IEEE Transactions on Green Communications and Networking* [In Revision].
- R. Gulia et al., "Sensing-Aware Backscatter Communications: A Survey on Envelope Stability, Waveform Design, and Selection Diversity," *Elsevier Ad Hoc Networks* (Under Review).

- R. Gulia et al., "Evaluation of 60 GHz Wireless Connectivity for an Automated Warehouse," *Information*, 2023.
- R. Gulia et al., "Automated Warehouse 5G Infrastructure Modeling Using VAEs," *ISNCC*, 2024.

EXPERIENCE

Research Lead | ARIX Lab (Independent Research Initiative)

Aug 2025 – Present

- **Lead collaborative research projects** in AI-native wireless systems, V2X, digital twins, and interpretable AI with academic and industry researchers.
- Mentor researchers and coordinate interdisciplinary projects targeting IEEE journal publications and emerging 6G applications.

Research Assistant | RIT

Sep 2020 – May 2026

- **Grant Proposal Development & Strategy:** Collaborated closely with **Dr. Amlan Ganguly** on the technical formulation and drafting of research proposals focusing on **AI-driven 5G/6G industrial architectures** for **Toyota University Research Program**. Assisted in identifying key research gaps, defining project milestones, and architecting the "Physics-Informed ML" workstreams that formed the core of funding requests.
- Designed and **optimized large-scale wireless network simulation pipelines** (ns-3 + ML), achieving 108× speedup and enabling real-time system-level tuning.
- Pioneered a physics-informed deep learning framework (WISVA) for accurate **predictive modeling of resource conditions** and **real-time spatial sensing** in complex, dynamic industrial environments.
- Developed a dynamic digital twin framework (**Evo-WISVA**) to model and predict spatiotemporal changes in wireless channels, achieving high accuracy (**MAE: 0.329 dB**).

ns3 Wireless Intern | AIRA TECHNOLOGIES, INC.

Jun 2023 – Aug 2023

- Conducted in-depth **performance analysis of an Open RAN deployment** using the ns-3 simulator, focusing on system-level performance.
- Collaborated on developing and enhancing **rApp functionalities** to improve network intelligence and scalability.
- **System-Level Performance Analysis:** Experience in evaluating Open RAN scalability and rApp logic within standardized networking frameworks (ns-3)

TEACHING EXPERIENCE & INTERESTS | RIT

August 2019 – May 2020

Course: EEEE 484 – Introduction to Communication Systems |

- Assisted in delivering core concepts to ~30 students
- Guided students on problem sets, labs, and exams
- Supported active learning and concept reinforcement

Technical Course Architect & Lead Instructor | Udemy

Jan 2026 – Present (Forthcoming)

- Curriculum Design: Architected a 2+ hour advanced technical course: **"Digital Communication Systems (Part 1): Information Theory & Noise,"** bridging the gap between abstract probability and physical-layer engineering.
- **Knowledge Synthesis:** Developed 10+ modules covering Gaussian Noise modeling, Shannon Capacity limits, and the mathematical transition from Analog to PCM (Pulse Code Modulation).

PROJECTS

4D Fresnel Space-Time Modulation (4D-FSM) for Near-Field 6G Systems

- Proposed a novel 4D communication framework **encoding information across angle, depth, synthetic velocity, and modulation symbols** in near-field ELAA systems. **Introduced kinematic multiplexing** via space-time modulation, enabling separation of co-located users through synthetic Doppler (velocity-orthogonal channels).
- Developed a Discrete Fresnel Transform (DFnT)-based precoding architecture, reducing complexity from $O(N^3)$ to $O(N \log N)$ and enabling real-time operation under high mobility.
- Demonstrated robustness at 140 GHz under extreme mobility (up to 200 m/s), maintaining high channel correlation where conventional beamforming collapses.

Backscatter-Aware Antenna Selection for Reliable Passive Sensors WSNs (IEEE 802.11bp/be)

- Designed and mathematically formulated the **Backscatter-Constrained Transmit Antenna Selection (BC-TAS)** framework to co-optimize primary link SNR, transmitter PAPR, and backscatter incident stability.
- Implemented a **Kalman-based channel smoothing** layer to maintain selection stability under 20% CSI estimation error, achieving a **25x improvement in energy efficiency** and a **2.4 dB reduction in PAPR** (allowing 3 dB lower PA back-off).
- Demonstrated a nearly three-order-of-magnitude reduction in outage probability, effectively bridging the "Illuminator's Gap" for hardware-linear, battery-free sensing coexistence.

Evo-WISVA: Dynamic Channel Digital Twin for Smart Warehouses

- Developed an AI/ML digital twin framework (**Evo-WISVA**) that predicts wireless channel conditions in dynamic warehouse environments achieving high accuracy (MAE=0.329 dB); achieved high accuracy and an average prediction speed of **0.5 msec per frame**.
- **High-Fidelity RF Channel Emulation:** Architected a Digital Twin framework to create standardized, repeatable test environments for 5G evaluation.

Stabilized Symbolic Regression for Interpretable BLER Modeling in V2X Systems (Collaboration with F. Papoola and Ashish Sheikh)

- Designed and implemented a physics-informed **Symbolic Regression** framework to discover explicit analytical expressions for V2X BLER, addressing the interpretability gap of black-box ML models.
- Utilized an **aggressive Genetic Programming** configuration (20,000 population, 300 generations) to derive a 253-node analytical expression with $R^2 = 0.86$ and $MSE = 2.08 \times 10^{-2}$ on unseen BLER data.

Modeling Air-Ground Channel for UAVs

(M.S. Thesis)

- **Conducted research to create a stochastic Air-Ground channel model** for UAVs in indoor environments, adaptable to 3GPP UAV-specific channels.
- **Proposed and experimentally validated a new path loss model (ULAAG)** based on RSSI measurements, considering UAV height, angle, and indoor characteristics.

TECHNICAL PRESENTATION

- **Evaluation of wireless connectivity in an automated warehouse at 60Ghz**, 2022 IEEE International Conference on Consumer Electronics (ICCE) 2022, Rochester, NY, 2022
- **Automated Warehouse 5G Infrastructure Modeling Using Variational Autoencoders**, 2024 International Symposium on Networks, Computers and Communications (ISNCC), Washington, DC, 2024

SERVICES & PEER RECOGNITION

- **Active Reviewer:** *Engineering Applications of Artificial Intelligence* (Elsevier)
 - Recognized with a **Certificate of Reviewing** for evaluating high-impact manuscripts in Deep Learning and Industrial AI (2025–2026).
- **Technical Committee Member:** IEEE Consumer Technology Society (CTSoc).
- **Peer Reviewer:** IEEE Internet of Things Journal, IEEE ICCE-Taiwan.
- **Research Leadership:** Led the wireless AI showcase at Imagine RIT, presenting complex research to diverse community and government stakeholders.

CERTIFICATIONS

- IBM Deep Learning & Reinforcement Learning (Coursera, May 2024)
- Generative Deep Learning with TensorFlow (Coursera, Mar 2024)
- VAEs for Image Compression (Coursera, Jan 2024)
- Neural Networks & Deep Learning (Coursera, Aug 2022)